

THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

Screening Test – Ramanujan Contest

NMTC at INTER LEVEL – XI & XII Standards

2025 – 2026

Instructions:

1. Fill in the response sheet with your Name, Class and the and the institution through which you appear in the specified places.
 2. Diagrams are only visual aids; they are NOT drawn to scale.
 3. You are free to do rough work on separate sheets.
 4. Use of Electronic gadgets such as Calculator, Mobile Phone or Computer is not permitted.
 5. Duration of the test: 10 am to 12 noon (Two hours).
 6. For each correct response you get 1 mark; for each incorrect response, you lose $1/2$ mark.
-

PART – A

Note

- Only one of the choices A, B, C, D is correct for each question. Shade the alphabet of your choice in the response sheet. If you have any doubt in the method of answering, seek the guidance of the supervisor.
- For each correct response you get 1 mark. **For each incorrect response you lose $\frac{1}{2}$ mark.**

1. If four different positive integers m, n, p, q satisfy the equation

$$(7 - m)(7 - n)(7 - p)(7 - q) = 4$$

then the sum $m + n + p + q$ is equal to

- A. 10 B. 24 C. 28 D. 36
2. A three member sequence a, b, c is said to be a *up-down* sequence if $a < b$ and $b > c$. For example 1, 3, 2 is a up-down sequence. The sequence 1 3 4 2 contains three up-down sequences: (1, 3, 2), (1, 4, 2) and (3, 4, 2). How many up-down sequences are contained in the sequence 1 3 2 5 9 7 6 8 4?

A. 32 B. 34 C. 36 D. 38

 3. For a positive integer n , let $P(n)$ denote the product of the digits of n when n is written in base 10. For example, $P(123) = 6$ and $P(788) = 448$. If N is the smallest positive integer such that $P(N) > 1000$, and N is written as $100x + y$ where x, y are integers with $0 \leq x, y < 100$, then $x + y$ equals

A. 112 B. 114 C. 116 D. 118

 4. The sum of 2025 consecutive odd integers is 2025^{2025} . The largest of these off numbers is

A. $2025^{2024} + 2024$ B. $2025^{2024} - 2024$ C. $2025^{2023} + 2024$
D. $2025^{2023} - 2024$

 5. ABC is an equilateral triangle with side length 6. P, Q, R are points on the sides AB, BC, CA respectively such that $AP = BQ = CR = 1$. The ratio of the area of the triangle ABC to the area of the triangle PQR is

A. 36 : 25 B. 12 : 5 C. 6 : 5 D. 12 : 7

6. How many three-digit positive integers are there if the digits are the side lengths of some isosceles or equilateral triangle?
- A. 45 B. 81 C. 165 D. 216

7. All the positive integers whose sum of digits is 7 are written in the increasing order. The first few are 7, 16, 25, 34, 43, ... What is the 125th number in this list?

- A. 7000 B. 10006 C. 10024 D. 10042

8. The bisectors of the angles A, B, C of the triangle ABC meet the circum circle of the triangle again at the points D, E, F respectively. What is the value of

$$\frac{AD \cos \frac{A}{2} + BE \cos \frac{B}{2} + CF \cos \frac{C}{2}}{\sin A + \sin B + \sin C}$$

if the circum radius of ABC is 1?

- A. 2 B. 4 C. 6 D. 8

9. For a real number x , let $[x]$ be the greatest integer less than or equal to x . For example, $[1.7] = 1$ and $[\sqrt{2}] = 1$. Let $N = \left\lfloor \frac{10^{93}}{10^{31} + 3} \right\rfloor$. Find the remainder when N is divided by 100.

- A. 1 B. 8 C. 22 D. 31

10. A point (x, y) in the plane is called a lattice point if both its coordinates x, y are integers. The number of lattice points that lie on the circle with center at $(199, 0)$ and radius 199 is

- A. 4 B. 8 C. 12 D. 16

11. The sum of all real numbers p such that the equation

$$5x^3 - 5(p+1)x^2 + (71p-1)x - (66p-1) = 0$$

has all its three roots positive integers.

- A. 70 B. 74 C. 76 D. 88

12. If $1 - x + x^2 - x^3 + \dots + x^{20}$ is rewritten in the form $a_0 + a_1(x-4) + a_2(x-4)^2 + \dots + a_{20}(x-4)^{20}$, where $a_0, a_1, \dots, a_{20}$ are all real numbers, the value of $a_0 + a_1 + a_2 + \dots + a_{20}$ is

- A. $\frac{5^{21} + 1}{6}$ B. $\frac{5^{21} - 1}{6}$ C. $\frac{5^{20} + 1}{6}$ D. $\frac{5^{20} - 1}{6}$

13. For a positive integer n , a *distinct 3-partition* of n is a triple (a, b, c) of positive integers such that $a < b < c$ and $a + b + c = n$. For example, $(1, 2, 4)$ is a distinct 3-partition of 7. The number of distinct 3-partitions of 15 is

- A. 10 B. 12 C. 13 D. 15

14. If m and n are positive integers such that $30mn - 6m - 5n = 2019$, what is the value of $30mn - 5m - 6n$?

- A. 1900 B. 2020 C. 1939 D. Can not be found from the given information

15. A class of 100 students takes a six question exam. For the first question, a student receives 1 point for answering correctly, -1 point for answering incorrectly or not answering at all. For the second question, the student receives 2 points for answering correctly and -2 points for answering incorrectly or not answering at all and so on. What is the minimum number of students having the same scores?

- A. 6 B. 5 C. 0 D. Can not be found from the given information

PART – B

Note

- Write the correct answer in the space provided in the response sheet.
- For each correct response you get 1 mark. **For each incorrect response you lose $\frac{1}{4}$ mark.**

16. The value of

$$\frac{1}{2} + \frac{1^2 + 2^2}{6} + \frac{1^2 + 2^2 + 3^2}{12} + \frac{1^2 + 2^2 + 3^2 + 4^2}{20} + \dots + \frac{1^2 + 2^2 + \dots + 60^2}{3660}$$

is _____.

17. The largest prime divisor of $3^{21} + 1$ is _____.

18. A circular garden divided into 10 equal sectors needs to be planted with flower plants that yield flowers of 3 different colors, in such a way that no two adjacent sectors will have flowers of the same color. The number of ways in which this can be done is _____.

19. We call an integer special if it is positive and we do not need to use the digit 0 to write it down in base 10. For example, 2126 is special whereas 2025 is not. The first 10 special numbers are 1, 2, 3, 4, 5, 6, 7, 8, 9, 11. The 2025th special number is _____.

20. Let a, b, c be non zero real numbers such that $a + b + c = 0$ and $a^3 + b^3 + c^3 = a^5 + b^5 + c^5$. The value of $\frac{5}{a^2 + b^2 + c^2}$ is _____.

21. The equation $x^3 - \frac{1}{x} = 4$ has two real roots α, β . The value of $(\alpha + \beta)^2$ is _____.

22. If x, y, z are positive integers satisfying the system of equations

$$xy + yz + zx = 2024$$

$$xyz + x + y + z = 2025$$

find $\max(x, y, z)$. _____.

23. If p, q, r are primes such that $pq + qr + rp = pqr - 2025$, find $p + q + r$. _____.

24. A cyclic quadrilateral has side lengths 3, 5, 5, 8 in this order. If R is its circumradius, find $3R^2$. _____.

25. Consider the sequence of numbers 24, 2534, 253534, 25353534, Let N be the first number in the sequence that is divisible by 99. Find the number of digits in the base 10 representation of N . _____.

26. An isosceles triangle has integer sides and has perimeter 16. Find the largest possible area of the triangle. _____.

27. Suppose that a, b, c are positive real numbers such that $a^2 + b^2 = c^2$ and $ab = c$. Find the value of

$$\frac{(a + b + c)(a - b + c)(a + b - c)(a - b - c)}{c^2}$$

_____.

28. In a right angled triangle with integer sides, the radius of the inscribed circle is 12. Compute the largest possible length of the hypotenuse. _____.

29. Points C and D lie on opposite sides of the line AB . Let M and N be the centroids of the triangles ABC and ABD respectively. If $AB = 25, BC = 24, AC = 7, AD = 20$ and $BD = 15$, find MN . _____.

30. Let $a_0 = 1$ and for $n \geq 1$, define $a_n = 3a_{n-1} + 1$. Find the remainder when a_{11} is divided by 97. _____.